

Model Type	Model	Climate Driver	AIC	Deviance Explained	R ² Adj	Climate term	Climate p-value (term) ¹	Significant contrasts vs. reference	EDF Terms	Inference
Punta Maroma										
Linear	M_the	Thermal regime	-34.71	0.565	0.418	Thermal	0.012	Thermal LIA (-1.81, p=0.003)	s(age) (EDF=2.53)	Supported
Smooth	M_hav	Hurricane average (events/100yr)	-59.19	0.834	0.703	s(Hur100_av)	<1e-04	n/a (no categorical terms)	s(age) (EDF=1); s(Hur100_av) (EDF=5.21)	Supported
Florida										
Linear	M_hyd	Hydroclimatic trends	-11.81	0.202	0.157	Hydro	0.12	none	s(age) (EDF=1)	Weak support
Smooth	M_hav	Hurricane average (events/100yr)	-11.11	0.216	0.100	s(Hur100_av)	0.25	n/a (no categorical terms)	s(age) (EDF=1.15); s(Hur100_av) (EDF=1.92)	Weak support
Belize										
Linear	M_the	Thermal regime	-16.62	0.646	0.437	Thermal	0.00099	ThermalCWP&MW A (-2.10, p=0.00137); ThermalLIA (-2.52, p=0.00029)	s(age) (EDF=2.37)	Supported
Smooth	M_hf	Hurricane frequency (events/100yr)	-18.55	0.731	0.482	s(Hur100_fqcy)	0.00057	n/a (no categorical terms)	s(age) (EDF=1); s(Hur100_fqcy) (EDF=6.16)	Supported
Saint Croix										
Linear	M_hur	Hurricane regime	-7.31	0.469	0.221	Hurricane	0.21	none	s(age) (EDF=1)	Weak support
Smooth	M_dt	thermal rate of change $\delta T(^{\circ}C)/100yr$	-7.44	0.522	0.303	s(dT_raw)	0.08	n/a (no categorical terms)	s(age) (EDF=2.11); s(dT_raw) (EDF=1)	Weak support
Regional										
Linear	M_the	Thermal regime	-25.71	0.364	0.202	Thermal	0.0019	ThermalLIA (-2.32, p=0.00019)	s(age) (EDF=2.56)	Supported
Smooth	M_hav	Hurricane average (events/100yr)	-23.02	0.302	0.136	s(Hur100_av)	0.024	n/a (no categorical terms)	s(age) (EDF=1); s(Hur100_av) (EDF=3.47)	Supported

¹Climate p-value: joint Wald test of the whole term (all factor levels together for categorical regimes; the spline for continuous predictors). Contrasts are relative to the reference levels warmer / wet / low.